

Report

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1 Introduction

This report fits a logistic regression model for the probability that a chess player wins an individual game (Reyes, 2025), using game-level attributes scraped from chess.com. Each observation is a single game with a binary 0/1 response, so the model is

$$y_i \sim \text{Bernoulli}(\pi_i), \quad \log\left(\frac{\pi_i}{1 - \pi_i}\right) = \mathbf{x}_i^\top \beta.$$

Model building proceeds in two stages. LASSO is first applied to screen a large pool of candidate terms, comprising main effects, quadratic terms, and all pairwise interactions down to a smaller active set. BIC bidirectional stepwise selection is then applied to that screened set to yield a parsimonious, unpenalised final model with valid coefficient estimates. Model adequacy is assessed using PIT residuals, which are the appropriate diagnostic tool for a discrete response because deviance and Pearson residuals cannot be expected to look Normal even under a correct model. The final model is applied to estimate the team’s winning probability against Women’s World Champion Ju Wenjun.

Beyond the technical sides, the practical value lies in its ability to quantify how much of a chess outcome is explained by measurable game attributes rather than by unobserved factors such as opening preparation or psychological pressure. By framing the final application as a concrete prediction problem against a named opponent, the report also makes the model’s conclusions easy to check against chess intuition.

Table 1: Group roster: students and their assigned chess.com players.

Student	Assigned Player	Username	Title	Country
Hongyu Zhou	Mohamed Anees M	THE_TITAN_15	IM	India
Mingzhuo Fan	Justin Lee	JLL2006	FM	USA
Zebo Hu	Simon Williams	Ginger_GM	GM	UK
Xinran Zhao	Tokhir Nematov	Brilliant_Move_11	CM	Uzbekistan

2 Data

The group roster is shown in Table 1. Each student is paired with a titled chess.com player whose game history is scraped for the analysis. The four assigned players are Mohamed Anees M (IM, `THE_TITAN_15`), Justin Lee (FM, `JLL2006`), Simon Williams (GM, `Ginger_GM`), and Tokhir Nematov (CM, `Brilliant_Move_11`).

Game data for all four players spanning 2024–2026 is retrieved from the chess.com public API. A helper function queries each player’s monthly PGN archive list, downloads every archive in the target years, and parses the PGN records using `read.pgn()`. The function handles failed requests and empty months gracefully via `try()`, and inserts a brief pause between API calls to avoid rate-limiting.

After combining the four players’ archives, the raw pool is filtered to retain only standard chess because the non-standard variants follow materially different rules and strategic logic. The chess.com API tags non-standard formats, including Chess960, 3-Check, Crazyhouse, and the unrated correspondence mode “Let’s Play” in the `Variant` PGN field and/or the `Event` string; any game matching these patterns is removed. Pooling such games would introduce heterogeneity that the model is not designed to capture and would bias the estimated effects toward whichever variant happened to dominate the sample. Of the 18,279 games scraped across all archives, 92 non-standard games are discarded at this stage, leaving 18,187 standard games.

Table 2: Win, draw and loss counts by assigned player (standard chess, 2024–2026).

Player	player wins	draw	player loses
Justin Lee	3970	631	3693
Mohamed Anees M	956	164	816
Simon Williams	3578	466	3213
Tokhir Nematov	328	28	344

The outcome breakdown by player appears in Table 2. Win rates differ meaningfully across players, reflecting their titles. Simon Williams (GM) posts the highest absolute win count, while the pattern across all four players confirms the dataset contains genuine variation in playing strength that the model can exploit (Elo, 1978).

Draws are excluded and only decisive games are retained (Hvattum, 2017), giving a clean binary response: `win` = 1 for a win, 0 for a loss. A draw is not an intermediate outcome on the win/loss scale but a qualitatively distinct result, often reflecting mutual agreement or a balanced position rather than a near win or a near loss (De Sa, 2012). Because draws make up only a modest share of decisive-versus-drawn games for these four players, their removal sacrifices little information while clarifying the response.

Rapid and Classical time controls are also removed, as they contain too few observations among the four assigned players to support reliable inference; the retained categories are Bullet, Bullet+increment, Blitz, and Blitz+increment. After filtering for complete Elo and movecount records, the modelling dataset comprises 16,896 decisive games with an overall win rate of 0.523 (Regan & Haworth, 2011).

Elo enters the model as the difference $\text{EloDiff} = \text{playerElo} - \text{opponentElo}$ (Fenner et al., 2011). Using the difference rather than the two ratings separately reflects the structure of the Elo system itself, in which only the gap between two players determines the expected score. The time control strings are collapsed into the four time categories above, and the player’s first move is extracted from the PGN move text and grouped into the five most frequent moves (separately for White and Black) plus an “Other” catch all.

Table 3: Exploratory summary of game outcomes across the modelling dataset, broken down by time control, playing colour, assigned player and first-move category. Each block reports game counts, win/loss split, win rate, and the mean Elo difference and game length for that subgroup.

	Games	Wins	Losses	Win rate	Mean EloDiff	Mean NMoves
Overall						
All games	16896	8831	8065	0.523	20.3	39.6
By time control						
Bullet	15342	7960	7382	0.519	10.0	39.5
Bullet+inc	1044	589	455	0.564	124.1	39.3
Blitz	390	211	179	0.541	92.1	43.5
Blitz+inc	120	71	49	0.592	192.2	42.2
By playing colour						
Black	8456	4268	4188	0.505	20.2	39.8
White	8440	4563	3877	0.541	20.3	39.4
By assigned player						
Justin Lee	7662	3969	3693	0.518	21.4	41.8
Mohamed Anees M	1772	956	816	0.540	26.5	40.0
Simon Williams	6790	3578	3212	0.527	23.9	36.9
Tokhir Nematov	672	328	344	0.488	-45.5	40.0
By first move						
d4	3835	2099	1736	0.547	12.6	38.8
e6	2640	1344	1296	0.509	15.9	40.0
b3	2466	1295	1171	0.525	8.0	41.3
Other	2390	1222	1168	0.511	43.8	38.4
d5	1292	644	648	0.498	8.7	42.7
c5	979	500	479	0.511	21.3	38.1
d6	852	444	408	0.521	39.0	39.6
Nf6	793	370	423	0.467	-1.7	39.3
e4	676	366	310	0.541	34.2	36.3
Nf3	622	348	274	0.559	50.1	41.3
f4	351	199	152	0.567	27.4	37.0

Table 4: Distribution of the numeric predictors in the modelling dataset.

Variable	Min	Q1	Median	Mean	Q3	Max	SD
EloDiff	-912	-31.2	15	20.3	63	2406	163.0
NMoves	0	27.0	37	39.6	50	311	17.8
playerElo	1804	2650.0	2717	2706.8	2790	3093	135.5
opponentElo	233	2617.0	2708	2686.5	2792	3357	205.0

3 Model Construction and Interpretation

3.1 LASSO Screening

LASSO is used here only as a screening device, not as the final estimator (Buhlmann & van de Geer, 2011). The L1 penalty shrinks coefficients toward zero, which is effective for discarding clearly irrelevant terms but also biases the surviving coefficients and invalidates the usual standard errors (Tibshirani, 1996). The choice of the cross validated lambda reflects the aim at this stage is to be inclusive and avoid discarding a term that the subsequent BIC step might value, accepting some false positives in exchange for a low false negative rate. Enforcing strong hierarchy after screening ensures that the active set remains interpretable, since an interaction is difficult to read without its parent main effects present.

The screening step solves the L_1 penalised logistic regression problem

$$\hat{\beta}^{\text{LASSO}} = \arg \min_{\beta} \left\{ -\frac{1}{n} \sum_{i=1}^n \left[y_i \mathbf{x}_i^\top \beta - \log(1 + e^{\mathbf{x}_i^\top \beta}) \right] + \lambda \sum_{j=1}^p |\beta_j| \right\},$$

where $\mathbf{x}_i$ stacks the full set of main effects, quadratic terms, and pairwise interactions, and $\lambda \geq 0$ is the penalty chosen by cross-validation.

The candidate pool contains six predictors (`EloDiff`, `isWhite`, `NMoves`, `TimeCategory`, `first_move_cat`, `player_fullname`), the two quadratic terms `EloDiff`² and `NMoves`², and all pairwise interactions. Expanding factors to dummies, this produces a design matrix with 16896 rows and 158 columns. An L_1 penalised logistic regression is fitted to this matrix with the penalty λ chosen by 10-fold cross validation. We have got $\lambda_{\min} = 8.54 \times 10^{-4}$ (the most permissive sensible choice, so that the screen does not prematurely discard terms that BIC may want), 95 of 158 design columns receive non-zero LASSO coefficients. Mapping these back to model terms and enforcing strong hierarchy. Any interaction retained implies both its parent main effects are also retained; leaves 21 of 23 candidate terms in the active set passed to stage two.

3.2 BIC Stepwise Selection

BIC is preferred over AIC at this stage because its heavier penalty, which grows with the logarithm of the sample size, favours a more parsimonious model. With nearly seventeen thousand observations the BIC penalty per parameter is substantial, so a term is retained only if it earns its place by a clear margin. This guards against the tendency of AIC to retain marginal predictors in large samples and produces a final model that is easier to interpret and less prone to overfitting.

Table 5: BIC comparison of the null, screened and final models.

Model	Coefficients	BIC
Null model (intercept only)	1	23397.82
LASSO-screened model	140	20929.87
Final model (BIC-stepped)	21	20102.45

An unpenalised logistic GLM is then fitted on the LASSO-screened active set, and bidirectional stepwise selection is run with the BIC penalty $k = \log(n)$ between the null (intercept-only) model and the screened model. The BIC values are reported in Table 4. The final model achieves a lower BIC than both the null and the LASSO-screened model, confirming that the two-stage procedure successfully identifies a parsimonious set. Note that the stage-two BIC values are descriptive comparisons among screened candidates only; because the screening used the same data, they do not constitute valid post-selection inference.

The retained terms define the following linear predictor for the log-odds of winning:

$$\begin{aligned} \log \frac{\hat{\pi}}{1 - \hat{\pi}} = & \beta_0 + \beta_1 \text{EloDiff} + \beta_2 \text{isWhite} + \beta_3 \text{NMoves} + \beta_4^\top \text{TimeCategory} \\ & + \beta_5^\top \text{player_fullname} + \beta_6 \text{NMoves}^2 \\ & + \beta_7 (\text{EloDiff} \times \text{NMoves}) + \beta_8^\top (\text{EloDiff} \times \text{TimeCategory}) \\ & + \beta_9^\top (\text{EloDiff} \times \text{player_fullname}) + \beta_{10}^\top (\text{NMoves} \times \text{player_fullname}), \end{aligned}$$

where bold β are vectors corresponding to the dummy variable contrasts of the respective factor. The model thus allows the Elo advantage effect to vary across time controls and players, and permits the movecount effect to differ by player and to enter non linearly through the NMoves^2 quadratic term.

3.3 Coefficient Estimates and Interpretation

Table 6: Estimated coefficients of the final Bernoulli logistic model (log-odds scale).

Term	Estimate	Std. error	z value	Pr(> z)
(Intercept)	-1.4864	0.0907	-16.3864	<0.0001
EloDiff	0.0149	0.0006	24.6073	<0.0001
isWhite	0.1565	0.0342	4.5779	<0.0001
NMoves	0.0401	0.0033	12.3107	<0.0001
TimeCategoryBullet+inc	-0.0897	0.0821	-1.0927	0.2745
TimeCategoryBlitz	-0.3464	0.1380	-2.5093	0.0121
TimeCategoryBlitz+inc	0.0143	0.2291	0.0626	0.9501
player_fullnameMohamed Anees M	0.3245	0.1592	2.0380	0.0415
player_fullnameSimon Williams	0.5863	0.0965	6.0750	<0.0001
player_fullnameTokhir Nematov	1.8788	0.2464	7.6246	<0.0001
I(NMoves ²)	-0.0001	0.0000	-4.7087	<0.0001
EloDiff:NMoves	-0.0001	0.0000	-7.4979	<0.0001
EloDiff:TimeCategoryBullet+inc	-0.0080	0.0005	-16.2017	<0.0001
EloDiff:TimeCategoryBlitz	-0.0057	0.0007	-7.9637	<0.0001
EloDiff:TimeCategoryBlitz+inc	-0.0103	0.0010	-10.2755	<0.0001
EloDiff:player_fullnameMohamed Anees M	0.0030	0.0007	4.1584	<0.0001
EloDiff:player_fullnameSimon Williams	0.0017	0.0005	3.4091	0.0007
EloDiff:player_fullnameTokhir Nematov	-0.0040	0.0008	-4.9614	<0.0001
NMoves:player_fullnameMohamed Anees M	-0.0078	0.0036	-2.1320	0.0330
NMoves:player_fullnameSimon Williams	-0.0152	0.0023	-6.6222	<0.0001
NMoves:player_fullnameTokhir Nematov	-0.0364	0.0056	-6.5556	<0.0001

The estimated coefficients on the log-odds scale are given in Table 5. The final model has a residual deviance of 1.9898×10^4 on 16875 degrees of freedom, compared with a null deviance of 2.33881×10^4 , indicating a meaningful improvement in fit over the intercept-only baseline. The AIC of the final model is 1.994×10^4 and the BIC is 2.01024×10^4 .

The odds-ratio(OR) in Table 6 facilitates interpretation. **EloDiff** carries an odds ratio above 1, confirming that each additional rating point of advantage raises the odds of winning, consistent with Elo theory. Other fitted effects also admit a direct chess interpretation. The positive **isWhite** coefficient corresponds to an odds ratio of roughly 1.17, meaning that, holding rating difference and all other terms fixed, moving first multiplies the odds of winning by about seventeen percent. This is a clean statistical confirmation of the long observed first move advantage (Seirawan & Silman, 1990). The **NMoves** main effect is positive while the **NMoves** squared term is negative, describing a relationship that rises and then gently flattens, where longer games are associated with a higher win probability up to a point, plausibly because the player has converted an advantage or survived an opening, but the benefit does not grow without limit.

The negative **EloDiff** by **TimeCategory** interactions indicate that the value of each rating point of advantage is smaller in increment and faster formats than in the Bullet baseline, consistent with the idea that faster or differently paced games introduce more variance and allow a lower rated opponent to draw level. The **EloDiff** by **player_fullname** interactions show that the rating advantage does not translate into wins at exactly the same rate for every player, capturing individual differences in how reliably each player converts a rating edge. The **NMoves** by **player_fullname** interactions similarly allow the move-count effect to differ by player, reflecting distinct playing styles across the four accounts.

Table 7: Odds ratios and 95% confidence intervals for the final model.

Term	Odds ratio	Lower 95% CI	Upper 95% CI
(Intercept)	0.2262	0.1891	0.2698
EloDiff	1.0151	1.0139	1.0163
isWhite	1.1694	1.0936	1.2504
NMoves	1.0409	1.0344	1.0476
TimeCategoryBullet+inc	0.9142	0.7773	1.0727
TimeCategoryBlitz	0.7073	0.5392	0.9269
TimeCategoryBlitz+inc	1.0144	0.6430	1.5841
player_fullnameMohamed Anees M	1.3833	1.0110	1.8875
player_fullnameSimon Williams	1.7973	1.4876	2.1717
player_fullnameTokhir Nematov	6.5456	4.0436	10.6344
I(NMoves ²)	0.9999	0.9998	0.9999
EloDiff:NMoves	0.9999	0.9999	0.9999
EloDiff:TimeCategoryBullet+inc	0.9921	0.9911	0.9930
EloDiff:TimeCategoryBlitz	0.9944	0.9930	0.9958
EloDiff:TimeCategoryBlitz+inc	0.9898	0.9879	0.9918
EloDiff:player_fullnameMohamed Anees M	1.0030	1.0016	1.0044
EloDiff:player_fullnameSimon Williams	1.0017	1.0007	1.0027
EloDiff:player_fullnameTokhir Nematov	0.9960	0.9945	0.9976
NMoves:player_fullnameMohamed Anees M	0.9923	0.9853	0.9994
NMoves:player_fullnameSimon Williams	0.9849	0.9805	0.9893
NMoves:player_fullnameTokhir Nematov	0.9642	0.9537	0.9748

Table 8: Likelihood ratio test comparing the null and final models.

Model	Residual deviance	Residual df	Deviance drop	df	p-value
Null model (intercept only)	23388.1	16895			
Final model (BIC)	19898	16875			
Likelihood ratio test			3490.1	20	<0.0001

3.4 Goodness of Fit

This test establishes that the final model improves on an intercept-only baseline. The overall fit of the final model relative to the null (intercept-only) model is assessed via a likelihood ratio test. The null model has a residual deviance of 2.33881×10^4 on 1.6895×10^4 degrees of freedom. The final model, which introduces `EloDiff`, `isWhite`, `NMoves`, `TimeCategory`, `player_fullname`, `NMoves2`, and the four interaction terms (`EloDiff`×`NMoves`, `EloDiff`×`TimeCategory`, `EloDiff`×`player_fullname`, and `NMoves`×`player_fullname`), reduces the residual deviance to 1.9898×10^4 on 1.6875×10^4 degrees of freedom.

The drop in deviance of 3490.1 across 20 degrees of freedom yields a p -value of $< 2.2 \times 10^{-16}$ under a χ^2_{20} reference distribution, providing overwhelming evidence that the final model fits the data substantially better than the null. The magnitude of the deviance reduction, over 3400 units across 20 parameters, indicating that the retained predictors, and in particular the Elo difference and its interactions, account for a large and statistically unambiguous portion of the variation in game outcomes. This is consistent with the theoretical role of Elo ratings as a measure of relative playing strength (Kovalchik, 2020). the model essentially recovers and refines the Elo prediction with additional structure from time control, player identity, and move count.

4 Diagnostics

4.1 PIT Residuals

For a Bernoulli response, deviance and Pearson residuals are themselves discrete and cannot be expected to follow a Normal distribution even under a correctly specified model. PIT residuals are therefore used. For each observation, a uniform draw is taken over $(0, 1 - \hat{\pi}_i)$ if $y_i = 0$ and over $(1 - \hat{\pi}_i, 1)$ if $y_i = 1$; the residual is then the standard Normal quantile of that draw. If the model is correctly specified, these residuals are i.i.d. standard Normal, making a Normal QQ-plot and residual scatter plots valid diagnostic tools.

The PIT residuals computed here have mean 0.0016 and standard deviation 1.0075, close to the theoretical values of 0 and 1 respectively. The interquartile range runs from -0.6757 to 0.6798, consistent with a standard Normal distribution. These summary figures give a first indication of adequate model specification.

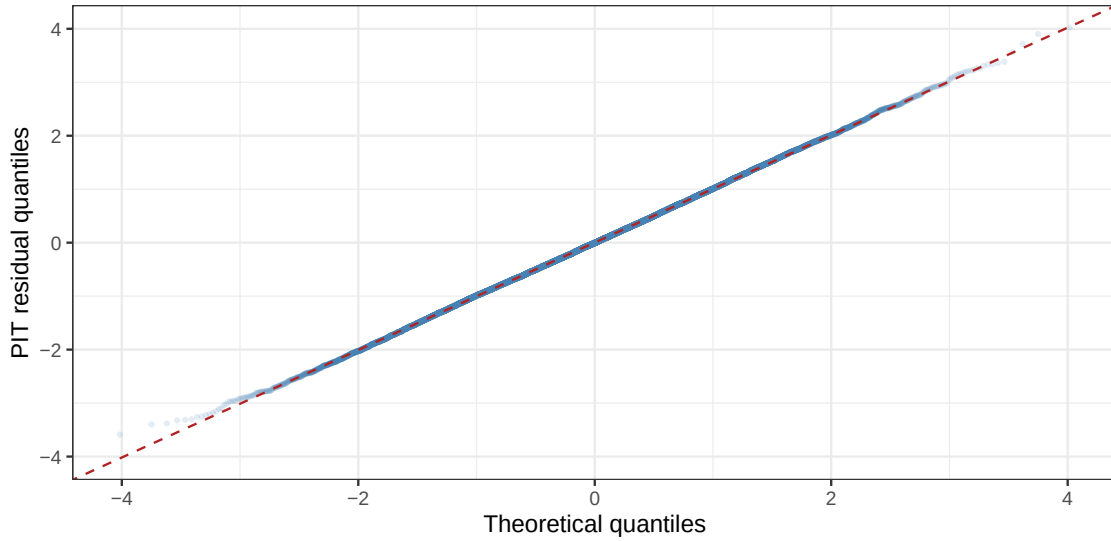


Figure 1: Normal QQ-plot of the randomised PIT residuals. Closeness to the dashed reference line indicates an adequately specified Bernoulli logistic model.

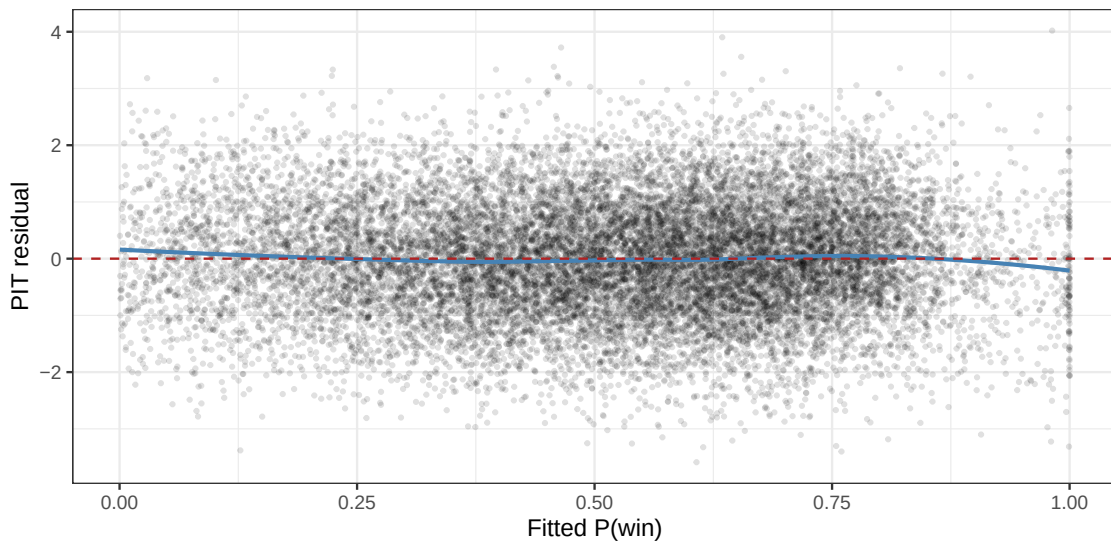


Figure 2: Randomised PIT residuals against fitted win probability. A loess smooth near zero indicates no systematic lack of fit.

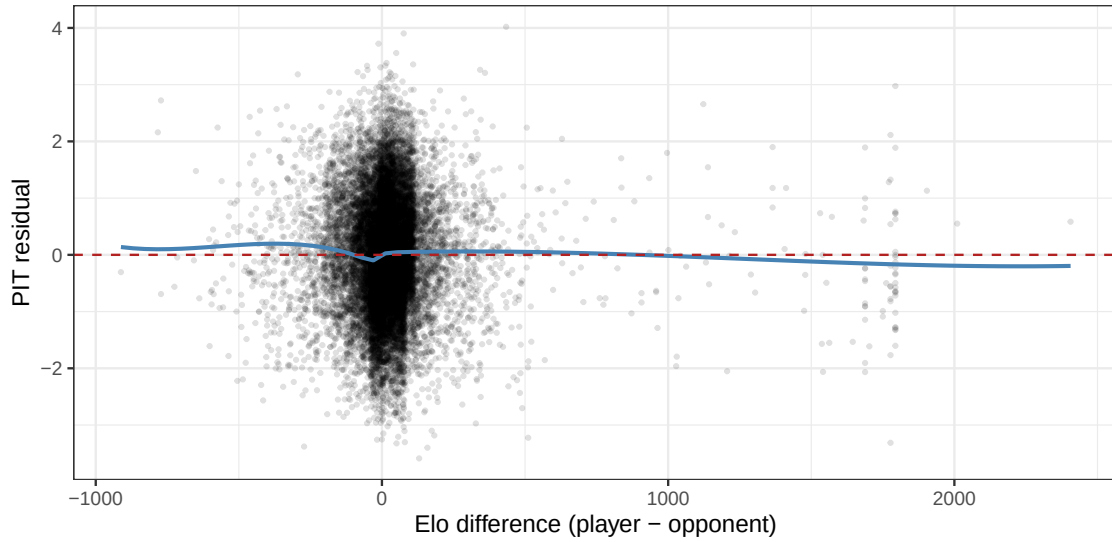


Figure 3: Randomised PIT residuals against Elo difference. A flat loess smooth indicates the Elo term is adequately modelled.

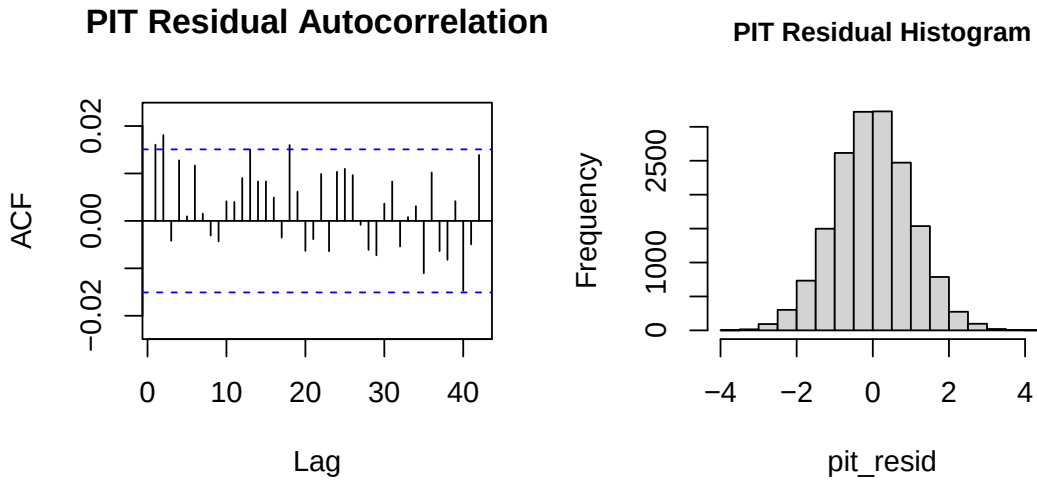


Figure 4: Autocorrelation function of the randomised PIT residuals. Bars within the dashed confidence band indicate no serial dependence.

The QQ-plot (Figure 1) shows points tracking closely along the reference line across the full range of quantiles, providing no strong evidence of misspecification. The residual scatter against fitted probability (Figure 2) shows the loess smooth lying near zero throughout, indicating no systematic pattern in the fit. Similarly, the scatter against `EloDiff` (Figure 3) reveals no curvature or trend in the smooth, supporting the adequacy of the linear Elo term. The histogram of PIT residual also shows no problem with model assumption. However, the ACF plot for PIT residual indicates that the independence assumption may slight off because there are some lags exceeds even after several lags but not severe. There may be some other patterns that our model not catch. For example, players will reflect and adjust their strategies after wins/losses, so a time series (eg. ARIMA) structure may be applied , but it is out of this courses' scope.

4.2 Influence Diagnostics

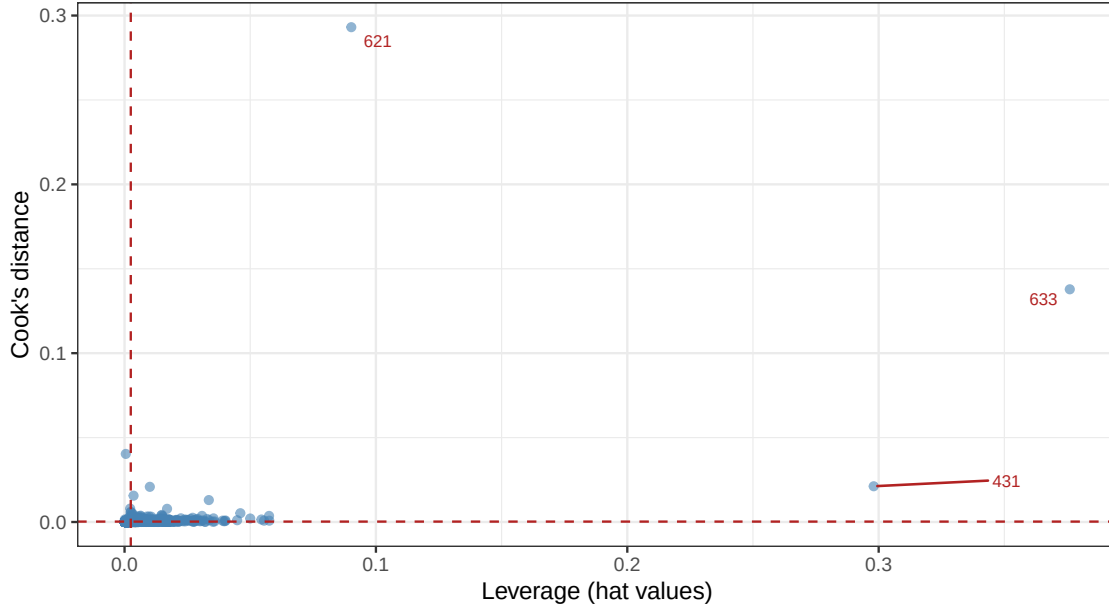


Figure 5: Leverage versus Cook’s distance. Dashed red lines mark the $2p/n$ and $4/n$ thresholds; flagged observations are labelled.

Leverage and Cook’s distance are used to identify observations with unusual predictor configurations or disproportionate influence on the fitted coefficients. The standard thresholds $2p/n = 0.00249$ for leverage and $4/n = 2.4 \times 10^{-4}$ for Cook’s distance flag 1875 observations in total (Figure 4). While some flagged points have non trivial influence by these criteria, their presence is not unexpected in a large dataset a small number of games with extreme Elo differences or unusual time-control configurations will naturally sit at the edge of the predictor space.

The three labelled observations in Figure 5 such as 633, with very high leverage, correspond to games with an unusual predictor configuration, may be an extreme rating gap or a rare time control combination, while point 621 combines moderate leverage with a large Cook’s distance, indicating a game whose outcome is poorly predicted given its attributes. Because no single observation among nearly seventeen thousand can materially distort the fitted coefficients, and because each flagged game appears to be legitimate, they are retained.

5 Applying the Model: Winning Chances Against Ju Wenjun

Ju Wenjun’s chess.com account (**Juwen**) is scraped using the same helper, subject to the same time control and variant filters as the main dataset. Only her games from 2025–2026 are used to represent her current playing strength, yielding 105 games with recorded Elo (from 224 scraped in total) (FIDE, 2026). Her mean Elo over this period is 2728. Because the four players have different titles and Elo levels, we retrieve the per-player mean Elo over 2025–2026 games. Aggregating these figures, the team Elo is 2666 under the mean. The mean is pulled upward by Simon Williams (GM, **Ginger_GM**), who has the highest Elo by a margin.

Table 9: Per-player Elo statistics over 2025–2026 games.

Player	Title	Mean Elo	Median Elo	Games
Mohamed Anees M	IM	2805	2808	1210
Simon Williams	GM	2737	2743	4839
Justin Lee	FM	2703	2709	7662
Tokhir Nematov	CM	2420	2466	672

Using the full scraped history would mix ratings from periods when Ju Wenjun was still consolidating her title

with her current peak performance, and would pull the representative Elo downward in a way that understates the difficulty of the matchup. Her mean Elo of 2728 places her in the top tier of active online players, well above the 2700 threshold that is widely regarded in the chess community as the boundary of grandmaster strength (FIDE, 2026). The proximity of her mean also suggests that her rating has been stable over this window rather than trending sharply in either direction, which supports treating 2728 as a reliable point estimate of her current strength. The mean aggregation gives a team Elo of 2666, implying a gap of roughly 62 rating points in Ju Wenjun’s favour.

Under the classical Elo formula, a 62 point deficit corresponds to a theoretical win probability of approximately 41 percent for the weaker side, treating the result as a continuous score rather than a decisive win (Glickman and Jones, 1999). The model will show that the choice of time control and playing color shifts the predicted probability by a non trivial amount around that baseline. Even a perfectly calibrated model assigns only a probability to any given game, and the realized result will always be binary. Narrow intervals, as seen for Bullet where the large number of training observations in that category pins the coefficients tightly, indicate that the model is confident about its estimate of the underlying probability. Wider intervals for Blitz and Blitz+increment reflect the smaller number of games in those categories among the four assigned players, and signal that predictions for those time controls carry more estimation uncertainty (Table 9).

Table 10: Sensitivity of predicted win probability (with 95% confidence interval) to time control and colour (mean Elo aggregation).

Time control	Black	White
Blitz	26.7% [21.3%, 32.9%]	29.9% [24.1%, 36.5%]
Blitz+inc	41.0% [29.4%, 53.7%]	44.8% [32.8%, 57.6%]
Bullet	26.6% [24.8%, 28.5%]	29.8% [27.9%, 31.8%]
Bullet+inc	35.2% [31.2%, 39.5%]	38.9% [34.6%, 43.3%]

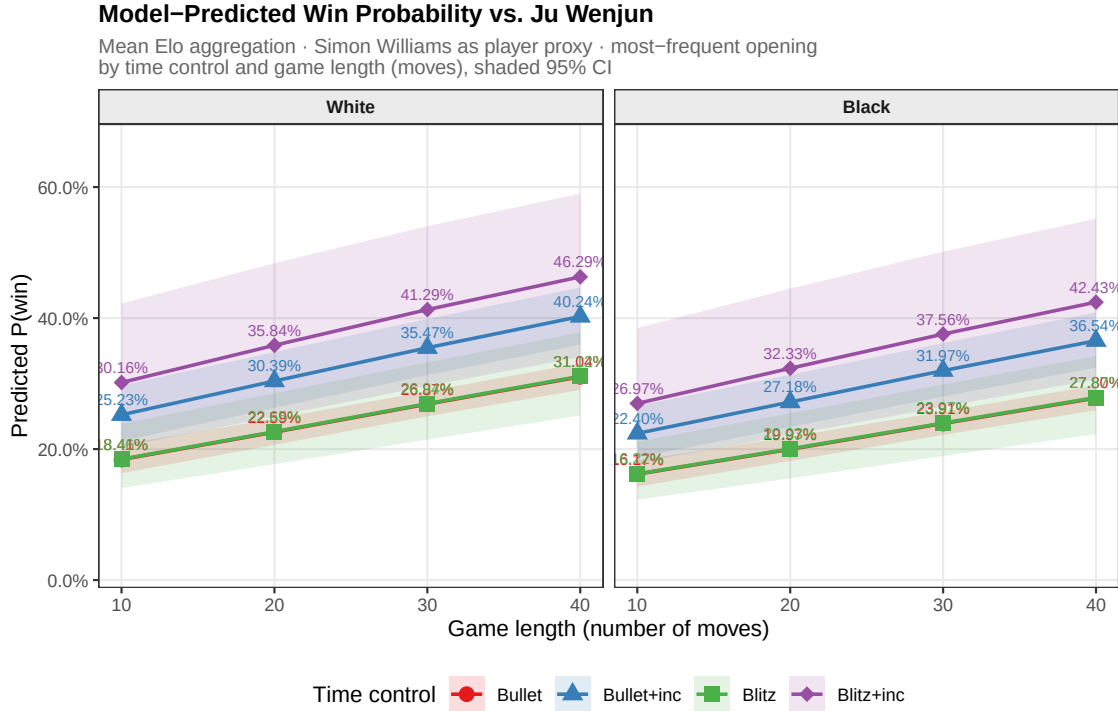


Figure 6: Model-predicted win probability against Ju Wenjun with 95% confidence bands, by time control and game length (mean Elo aggregation).

The sensitivity of these predictions to playing colour and time control, holding the Elo difference fixed at the mean aggregated value, is reported in Table 9. Playing as White consistently raises the predicted win probability relative to Black across all time controls, reflecting the first move advantage captured by the `isWhite` coefficient. Variation

across time controls is more modest but present, indicating that the time category effects retained by BIC do contribute to the prediction.

The predictions is consistent with section 3. The team faces a structural disadvantage at every time control and from either color, with win probabilities ranging from roughly 17 percent in the most unfavorable conditions (Bullet as Black at short game lengths) to just over 46 percent in the most favourable (Blitz+increment as White at longer game lengths). The spread across time controls within a fixed colour is larger than the White versus Black gap within a fixed time control. If the team were given a choice of format, Blitz+increment as White offers the highest predicted win probability and the widest confidence interval, meaning both the best expected outcome and the most variance, which is the rational preference for the underdog side (Glickman and Jones, 1999).

Longer games tend to favour the team, consistent with the positive NMoves coefficient in the model, which may reflect the fact that stronger opponents are more dangerous in short tactical encounters than in endgames that reward positional precision. This pattern holds across all four time controls, suggesting it is a genuine feature of how the assigned players perform rather than an artefact.

6 Limitations and Conclusions

Game data for the four assigned chess.com players was scraped across 2024–2026, combined into a single dataset, and filtered to retain only standard decisive games in Bullet, Bullet+increment, Blitz, and Blitz+increment time controls. A logistic regression model was built through a two stage procedure: LASSO screening of a large candidate pool followed by BIC bidirectional stepwise selection on the screened active set. PIT residuals confirmed adequate model specification except slight mismatch of independence, whereas the QQ plot and residual scatter plots showing no systematic departures from the standard Normal benchmark. Influence diagnostics identified a small number of high leverage or high Cook’s distance observations, consistent with the presence of games at extreme Elo differences in a large dataset. The model was applied to a hypothetical match against Ju Wenjun. The model predicted win probability is very low and aligns closely with the theoretical Elo expected score, providing a coherent consistent assessment.

There are a few limitations. The response was restricted to decisive games, so the model speaks to win probability conditional on a decisive result rather than to the full threeway outcome, and an ordinal or multinomial extension that models draws explicitly would give a more complete picture. The team Elo is an aggregate of four distinct players rather than the rating of a single competitor, so the prediction against Ju Wenjun is best read as an indicative benchmark rather than a precise forecast. Finally, the postselection coefficient estimates should be interpreted with mild caution, since the same data informed both the screening and the selection stages. Even so, the consistency between the model output, Elo theory, and chess intuition gives reasonable confidence that the model captures the main drivers of game outcomes for these players.

7 Reference

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8 Appendix

The analysis relies on `rjson` and `bigchess` for data retrieval and PGN parsing, `glmnet` for LASSO, `tidyverse` for data manipulation, and `knitr` with `kableExtra` for presentation. Model output is tidied with `broom`.